EEE8089
M2M Technology IoT
Topic 1: Introduction

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Aims

- Knowledge of Internet-based system interconnection technology or M2M:
  - networking,
  - system frameworks,
  - commercial examples.
- Skills in implementing an M2M network.

Intended learning outcomes

- OSI reference model, protocols, packet formats
- LAN, a combination of LANs with packet routing
- SSH in detail, high levels of OSI, security
- M2M architectures, DeviceHive, Google Cloud IoT, Amazon AWS, SSH ad-hoc
- Constructing an ad-hoc M2M network with SSH
- System design and implementation of a real-time control system with M2M
- Programming a Linux-based embedded system, interrupts, scripts, configurations, C, networking, debugging.

Rationale and Assessment

The formal lectures cover the theory:

- networking aspect: the OSI model with multiple layers, their implementations and formats;
- embedded systems aspect: a simple control system with the corresponding low-level interfaces;
- a combination of the above to form a M2M network.

Practicals develop skills and help to understand deeper the theory:

- Software design – to implement in software a control system interfaced to electro-mechanical hardware by using the Internet.

Assessment: 80% report, 20% presentations of the practical design.

Useful reading

- J. F. Kurose, K. W. Ross, Computer Networking: A Top-Down Approach, Global Edition [Book], 7th edition, Pearson, 2016 (earlier editions are suitable)
- Andrew S. Tanenbaum, David Wetherall, Computer Networks [Book], Pearson Prentice Hall, 2011 (earlier editions are suitable)
- Lydia Parziale, Dr. Wei Liu, Carolyn Matthews, Nicolas Rosselot, Chuck Davis, Jason Forrester, David T. Britt, TCP/IP Tutorial and Technical Overview [Free eBook], IBM Redbooks, 2006
- https://aws.amazon.com/iot-core/
- https://devicehive.com/
- https://cloud.google.com/solutions/iot/
- https://www.ssh.com/ssh/protocol/

What is IoT

- Often a political concept, a technology that potentially may make its users dependent on some form of authority, national or global – expect interference of governments, EU, UN, etc.
- Either humour or sophistry on Wikipedia: “...creating opportunities for more direct integration of the physical world into computer-based systems...” (a desire of impossible a.k.a. madness). Confusing, isn’t it?
- For definitions from see, for example, the following
  - Vermesan, Ovidiu; Friess, Peter, Internet of Things: Converging Technologies for Smart Environments and Integrated Ecosystems, Aalborg, Denmark: River Publishers, 2013.

IoT definition

Ovidiu Vermesan et al. (EC):

“Objects make themselves recognizable and they obtain intelligence by making or enabling context related decisions thanks to the fact that they can communicate information about themselves. They can access information that has been aggregated by other things, or they can be components of complex services”.

- Smart everything: cities, energy, transportation, factories, manufacturing, health, food, water, “participatory sensing” (control over personal association), social networks, etc.
- Privacy and security are important!
- Engineering perspective: interoperation of the “things” are at the levels of the OSI above TCP/IP.

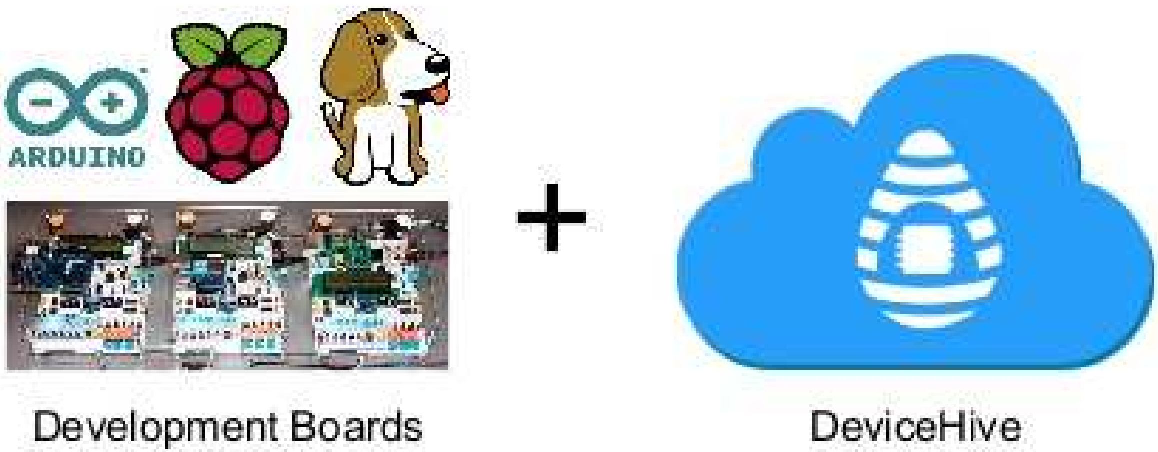
M2M

Engineer's definition [AB]:
A system of hardware devices, processes and services interconnected by means of the Internet at the presentation level of the OSI reference model.

- Let's name such a framework as *M2M Technology*
- The actual Internet is "hidden" from the system by means of the framework that provides an abstraction for the means of communication.
- A framework of IoT is represented as Internet-connected programs running in the devices and providing API to the other programs.

M2M: DeviceHive idea

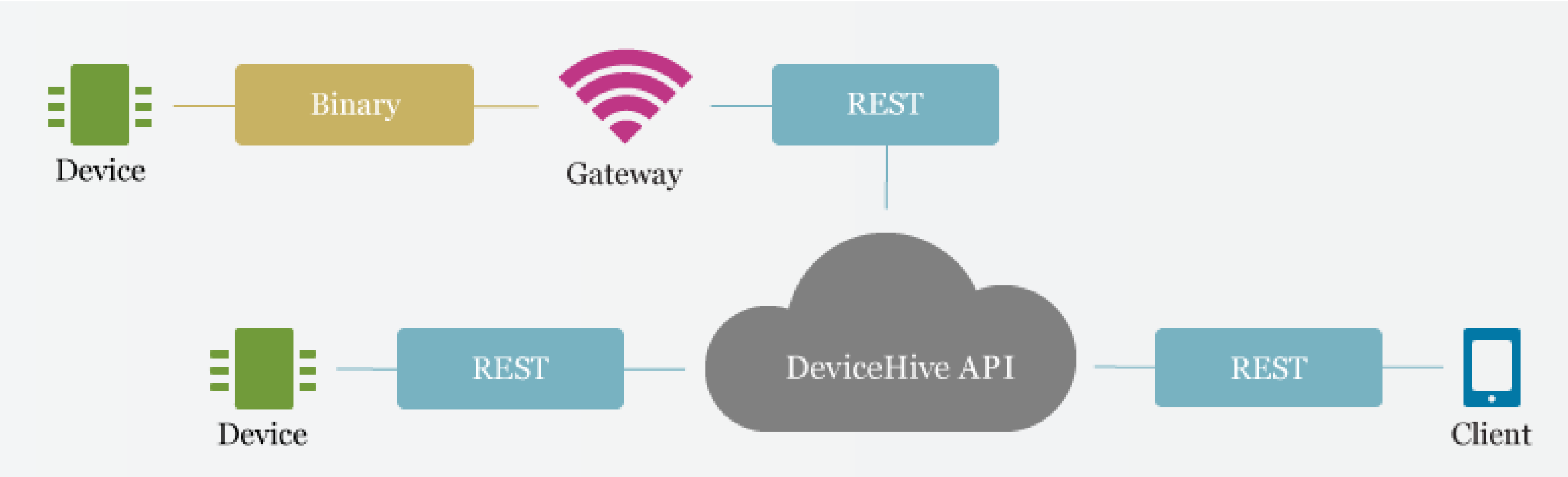
Solution



- No need to bother with low-level electronics
- Code in any language you are comfortable with
- Hassle-free M2M logic implementation with DH API libraries
- Quick DeviceHive server instance deployment

M2M: DeviceHive architecture

Open source, MIT public license



https://www.dataart.com/
https://devicehive.com/

M2M: DeviceHive “things” (1)

- Device:
  - communicates with the API of the framework,
  - a unit that runs microcode,
  - register – send notifications – poll commands.
- Client:
  - communicates with the API of the framework,
  - an application than monitors/controls the devices,
  - view devices – poll notifications – send commands.

M2M: DeviceHive “things” (2)

- Administrator:
  - communicates with the API of the framework,
  - manage users – networks – clients – devices.
- Framework:
  - provides the API to the above
  - a set of applications run on multiple Internet machines connected to devices or executing clients/administrators,
  - the only M2M component directly aware on the Internet.

M2M: DeviceHive connections (1)

- Device – framework:
  - the device is represented as a its driver in the M2M gateway machine, the actual interface can be RS232, Bluetooth, I2C, etc.,
  - we develop a “connector”, i.e. a program that connects the driver to the framework.
- Client/Administrator – framework:
  - the framework is represented as the API of the framework process running in a M2M gateway, it uses “REST” language to communicate with the framework central server,
  - we develop code that communicates to the whole system via the API.

M2M: DeviceHive connections (2)

- Framework – framework:
  - M2M gateways are talking to the M2M server in REST language.
- M2M server controls the framework and has a web interface to various data processing/visualisation tools.

Please visit DeviceHive web site to see their examples.

https://devicehive.com/

Other M2M platforms – home task

- Please find on the Internet any other M2M platforms
- Google
- Amazon
- Review and compare their general characteristics

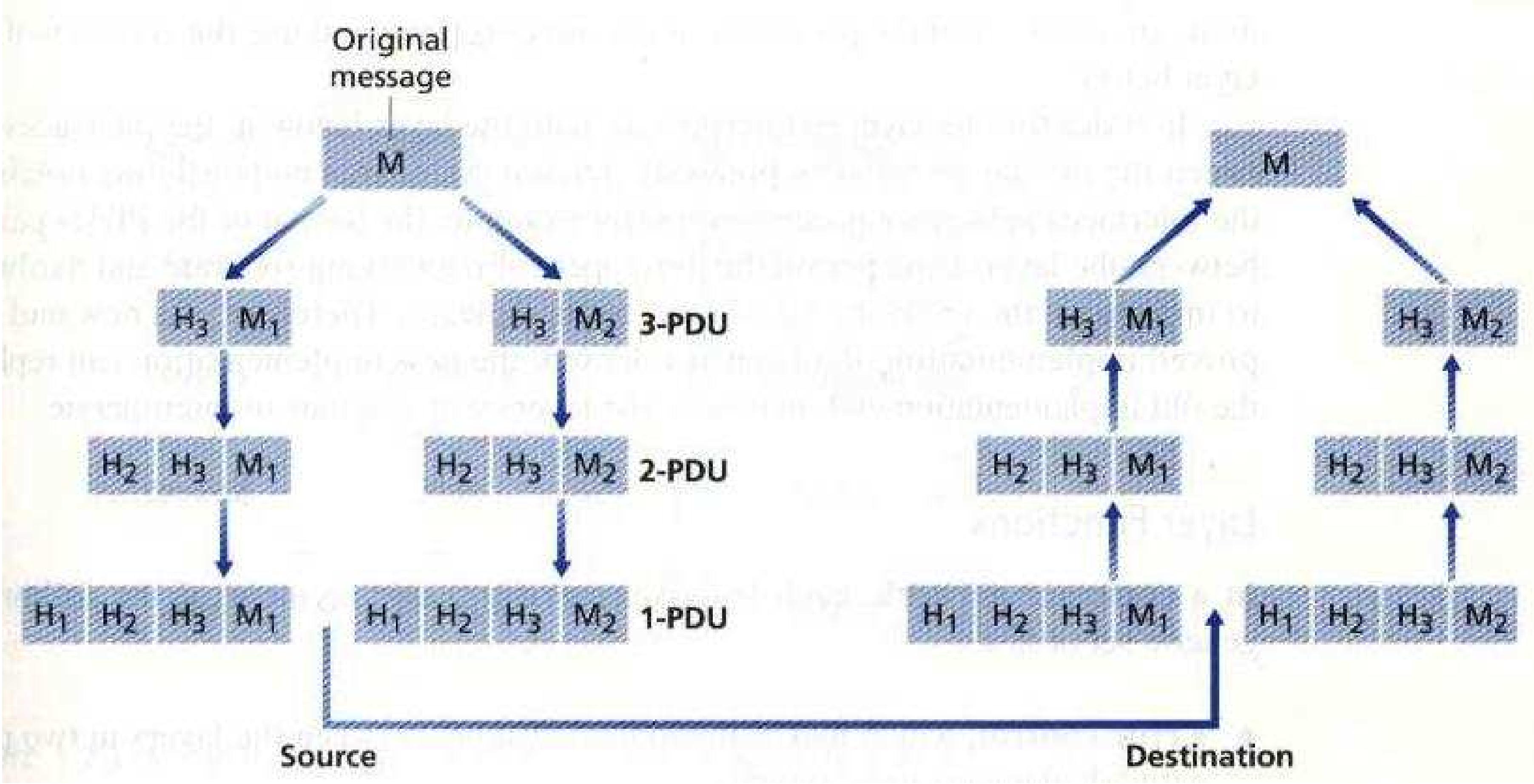
Background

- Protocol encapsulation
- OSI reference model
- Identify the level where the Internet ends and M2M begins
- Internet basics

Layered protocol

- Each layer builds upon a layer below;
  - adding new functionality.
- The lowest level
  - specific network hardware
  - no knowledge about the order of packets.
- The top level(s)
  - file transfer
  - mail delivery, etc.
- Benefits
  - new protocol development
  - transfer to a new hardware

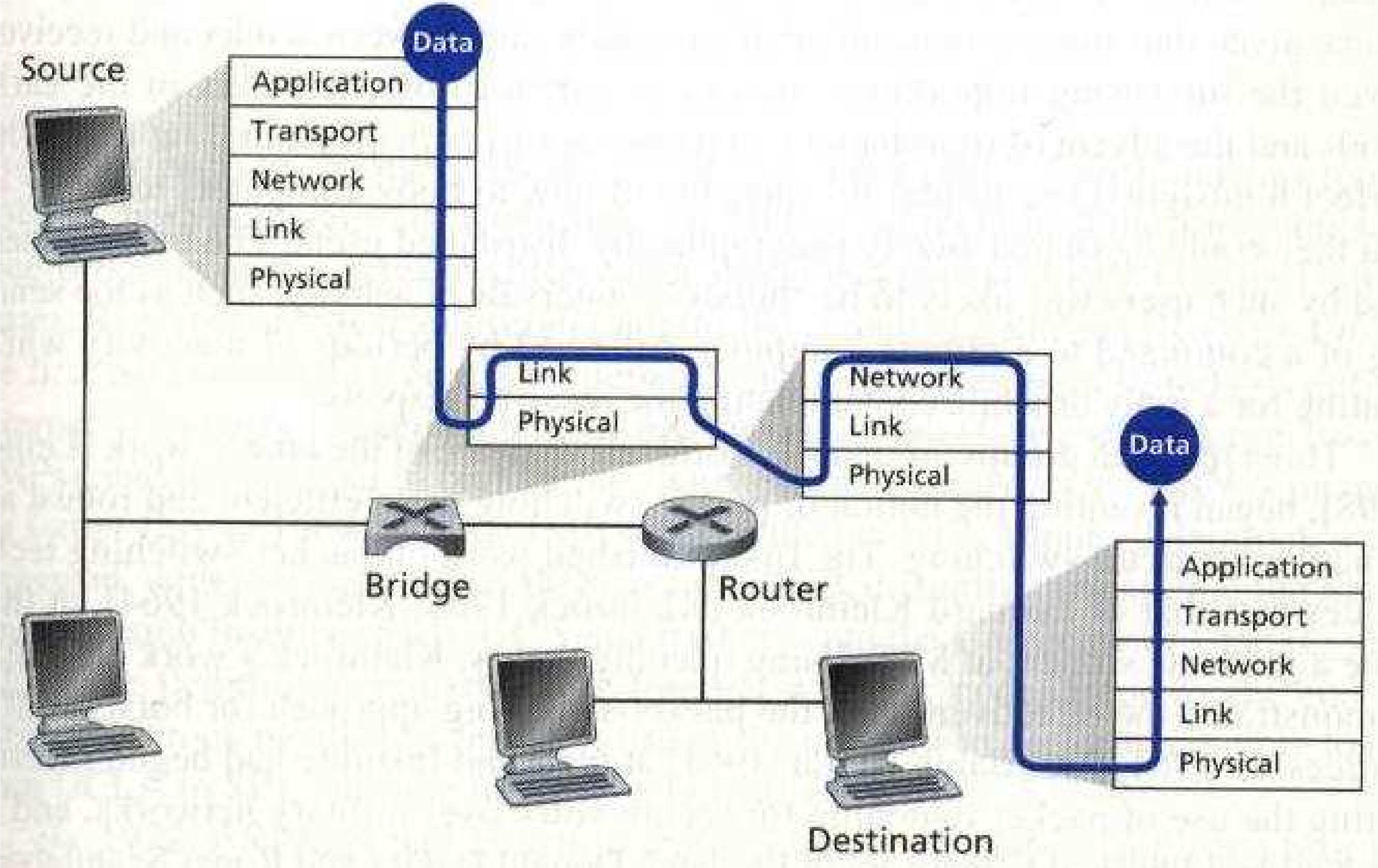
Protocol Data Units (PDU)



Protocol encapsulation principle.

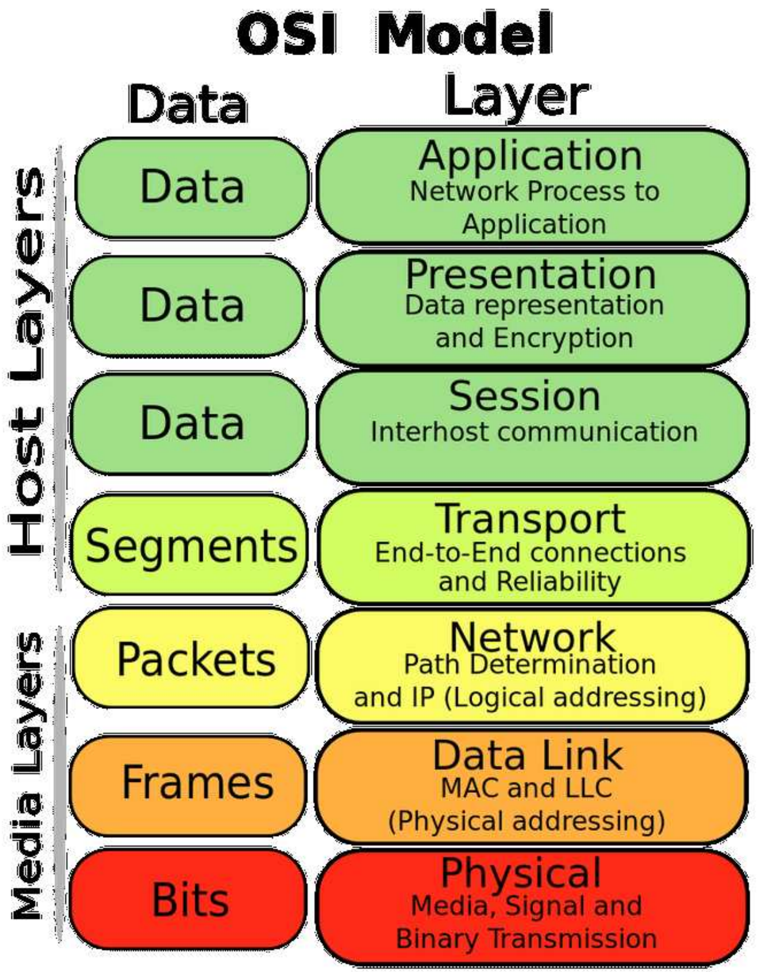
J. F. Kurose, K. W. Ross: “Computer Networking”, 2nd edition, Addison Wesley, 2003.

Network entities and layers



J. F. Kurose, K. W. Ross: “Computer Networking”, 2nd edition, Addison Wesley, 2003.

OSI diagram


OSI Model
Data
Layer
Host Layers
Media Layers
Data
Application
Network Process to Application
Data
Presentation
Data representation and Encryption
Data
Session
Interhost communication
Segments
Transport
End-to-End connections and Reliability
Packets
Network
Path Determination and IP (Logical addressing)
Frames
Data Link
MAC and LLC (Physical addressing)
Bits
Physical
Media, Signal and Binary Transmission

Application (Layer 7)

This layer supports application and end-user processes. Communication partners are identified, quality of service is identified, user authentication and privacy are considered, and any constraints on data syntax are identified. Everything at this layer is application-specific. This layer provides application services for file transfers, e-mail, and other network software services. Telnet and FTP are applications that exist entirely in the application level. Tiered application architectures are part of this layer.

Presentation (Layer 6)

A better name could be REPRESENTATION
This layer provides independence from differences in data representation (e.g., encryption) by translating from application to network format, and vice versa. The presentation layer works to transform data into the form that the application layer can accept. This layer formats and encrypts data to be sent across a network, providing freedom from compatibility problems. It is sometimes called the syntax layer.

Session (Layer 5)

This layer establishes, manages and terminates connections between applications. The session layer sets up, coordinates, and terminates conversations, exchanges, and dialogues between the applications at each end. It deals with session and connection coordination.

Transport (Layer 4)

This layer provides transparent transfer of data between end systems, or hosts, and is responsible for end-to-end error recovery and flow control. It ensures complete data transfer.

Network (Layer 3)

This layer provides switching and routing technologies, creating logical paths, known as virtual circuits, for transmitting data from node to node. Routing and forwarding are functions of this layer, as well as addressing, inter-networking, error handling, congestion control and packet sequencing.

Data Link (Layer 2)

At this layer, data packets are encoded and decoded into bits. It furnishes transmission protocol knowledge and management and handles errors in the physical layer, flow control and frame synchronisation. The data link layer is divided into two sub-layers: The Media Access Control (MAC) layer and the Logical Link Control (LLC) layer. The MAC sublayer controls how a computer on the network gains access to the data and permission to transmit it. The LLC layer controls frame synchronisation, flow control and error checking.

Physical (Layer 1)

This layer conveys the bit stream - electrical impulse, light or radio signal – through the network at the electrical and mechanical level. It provides the hardware means of sending and receiving data on a carrier, including defining cables, cards and physical aspects. Fast Ethernet, RS232, and ATM are protocols with physical layer components.

Identifying M2M in OSI

- The first 4 OSI levels are "just the Internet", used inside M2M frameworks.
- Level 5 "Session" is where the "things" become registered to M2M framework.
- Level 6 "Presentation" is the API provided by the framework – the devices/clients/administrators use this API.
- Level 7 "Application" is where the system belongs with its topology, protocols and languages. It is unaware of the method of communication (Internet or OSI 1-4) and the method of registration of the "things" (OSI 5). It uses the OSI 6 API for communication.

Conclusions

- M2M definition.
- Relation of M2M to OSI model.
- A general idea of M2M.
- Reinforcing with an example, DeviceHive.
- The necessary background on protocol encapsulation and PDUs.

What's next?

- The necessary background on the Internet, the lower 4 levels of OSI.
- This is necessary for M2M framework implementation.
- Then we'll construct several overlapping LANs to run M2M across them.